
Knowing Before Saying: Internal Correctness Signals in Mathematical Reasoning

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Abstract

1 Language models often answer math problems wrong while reporting high
2 confidence. Does the model lack self-knowledge, or does it have it and fail to
3 express it? To separate the two, we read the model’s internal state directly
4 instead of asking it. We test four open models with distinct architectures
5 on four domains: MMLU-Pro, MATH, GPQA-Diamond, and a geometry-
6 construction task whose outputs a symbolic compiler grades exactly. A
7 simple linear detector trained on that state, a probe, predicts whether an
8 attempt is correct better than the model’s stated confidence in 12 of 16
9 cells (one tie, mean +0.09 AUROC). The gap is largest on MATH (+0.20),
10 where a wrong answer looks as clean as a right one. The signal is specific to
11 the attempt: it survives a within-question control that pushes the model’s
12 own $P(\text{True})$ toward chance. It is substantially shared across domains: a
13 probe trained in one domain still discriminates in the others. And it causally
14 influences what the model says. On Mistral, amplifying a correctness
15 direction that was learned from labeled examples and then applied without
16 per-example labels at intervention time drops stated confidence on wrong
17 answers from 84 to 55 while correct answers hold near 90, and cuts calibration
18 error from 0.33 to 0.18. Models carry substantially more information about
19 their own errors than they state; we read that as knowing, and test the
20 reading causally. How much of that knowledge reaches speech varies by
21 architecture, and we outline a global-workspace test of why.

22 1 Introduction

23 A mathematical agent that errs is unavoidable. One that errs *and knows it* can abstain, retry,
24 or escalate. Stated confidence is a weak guide to correctness, and weakest where mathematics
25 is hardest. A wrong multi-step derivation still ends in a cleanly formatted final answer, and
26 nothing on the page gives it away. Section A walks one such attempt end to end: a single
27 wrong subtraction inside an otherwise correct derivation, after which the model *raises* its
28 stated confidence.

29 We split the problem into two questions. Do models internally represent that an output is
30 wrong (**knowing**)? Does that representation reach what they report (**saying**)? If a gap
31 exists, it is both an unused reliability signal and a concrete claim about machine introspection.

32 Prior work shows that output probabilities are often better calibrated than words [4], and
33 that the truth of *given statements* is linearly readable from activations [1, 2, 6]. Verbalized-
34 calibration studies measure what models say [5, 7]. We combine these threads on the model’s

own attempts. We elicit confidence before and after each attempt, measure blind (no-feedback) updating, take four readouts of the same latent quantity on identical records, and intervene on the representation [8]. “The model knows” is often used for five different claims at once, and our experiments separate them (Section 2). *Representation*: correctness is linearly decodable from internal state. *Specificity*: what is decoded tracks this attempt, not the difficulty of the question. *Generality*: one direction reads across domains. *Causal access*: manipulating it changes the confidence the model reports. *Expression*: stated confidence exposes only part of it, by an amount that depends on architecture. Decoding alone establishes only the first, which is why we test the other four; *knowing* is our reading of the five together.

2 Knowing versus saying

Models and domains. We test four open models with different architectures: Qwen3.6-27B (hybrid linear/full attention), GLM-4.7-Flash (MoE), Mistral-Small-24B (dense), and Gemma-4-26B-A4B (MoE, VLM-derived). Each runs on four domains, about 750 records per cell: MMLU-Pro, MATH, and GPQA-Diamond with programmatic grading, and a geometry-construction task (GeoGenBench) in which the model writes a construction from a natural-language request and a symbolic compiler decides each clause of the request exactly. The compiler matters because metacognition experiments live or die by their correctness labels, and it grades exactly against the formal specification: a pre-registered human study ($N=200$, two trained raters) found its disagreements with humans ran in one direction only, lenient and never harsh, so a **fail** label is trustworthy. Geometry is also the one generative domain; the others are answer-selection or short-answer tasks. We dropped benchmarks the models have saturated, since self-knowledge of failure cannot be studied where there are no failures.

Protocol. Each problem takes three turns: prospective confidence, the attempt, and retrospective confidence. Grading is external and never shown to the model, so any post-attempt revision is blind self-assessment. We take four readouts of the same latent quantity on identical records: stated confidence (0–100); $P(\text{True})$, the model’s probability of answering “True” when asked whether its answer is correct; the answer’s log-probability; and a linear *probe*. The residual stream is the model’s internal state, a vector of a few thousand numbers at every token and layer. The probe is a logistic regression trained to read correctness from that vector at the confidence decision token, the position that is about to produce the confidence digit. It is scored out-of-fold with question-grouped folds and reported at one pre-specified layer at 0.7 of network depth rather than the best layer. Our metric throughout is AUROC, the probability that a random correct attempt scores above a random wrong one: 0.5 is chance, 1.0 is perfect. Four controls guard the readout. The probe adds +0.22 mean AUROC over a surface-features baseline (answer length and shape) in 15 of 16 cells. The read token is identical across records, so layer-0 activations carry no signal by construction (per-fold AUROC 0.500); anything decoded deeper was computed during the attempt. A within-question control holds difficulty fixed. A public deviations log records four bugs that planned controls caught, which we fixed and re-ran.

Representation: knowing exceeds saying, most where errors look clean. The internal state ranks right and wrong answers better than the model’s own words do. The probe beats stated confidence in **12 of 16** cells (Table 1, left). The three losses are small- n cells, two of them Gemma-4. The gap is widest on MATH (mean +0.20). Stated confidence there is weak, while the internal signal is the strongest of any domain (up to 0.95). Verbalized confidence appears to lean on surface fluency, and multi-step computation defeats fluency. Post-attempt internal readouts also exceed pre-attempt ones. Doing the work sharpens the internal signal even where it degrades the stated one.

Specificity: the signal tracks the attempt, not the question. A skeptic’s alternative is that the probe has only learned which questions are hard. Holding the question fixed rules this out. Within a single question, across its five sampled attempts, only attempt-level information can discriminate. There the model’s own $P(\text{True})$ falls toward chance while the probe holds (Table 1, right). $P(\text{True})$ largely recovers *which questions are hard*. The probe

Table 1: Left: post-attempt correctness discrimination (AUROC), stated confidence vs. residual-stream probe, in representative cells. Overall the probe wins 12 of 16 cells with one tie, mean +0.09. A paired bootstrap is significantly positive in 10 of 16 and never significantly negative, uncorrected at the cell level (Section B). Right: the within-question control, with difficulty fixed. The model’s own P(True) falls toward chance while the probe holds.

model $\times$ domain	stated	probe
Gemma-4 $\times$ MATH	0.57	0.78
Qwen3.6 $\times$ MMLU-Pro	0.67	0.84
GLM $\times$ MMLU-Pro	0.67	0.85
Mistral $\times$ MATH	0.79	0.83

within-question	probe	P(True)
Qwen3.6 $\times$ MMLU-Pro	0.74	0.47
GLM $\times$ MMLU-Pro	0.73	0.59
Mistral $\times$ MATH	0.72	0.59

Table 2: Left: cross-domain probe transfer for Mistral on a fresh 2,614-record capture (rows train, columns test; AUROC at the fixed layer; diagonal is in-domain out-of-fold; Geo = GeoGenBench). Right: steering Mistral on MATH by amplifying its own correctness direction at the confidence token, with the random control scaled to the same per-record perturbation magnitude (stated confidence on failed / correct answers; ECE).

train $\downarrow$ test $\rightarrow$	Geo	MMLU-Pro	MATH	GPQA
Geo	0.68	0.67	0.78	0.56
MMLU-Pro	0.70	0.75	0.84	0.59
MATH	0.80	0.72	0.83	0.57
GPQA	0.60	0.69	0.74	0.57

gain	1	2	4
steer, fail / ok	84 / 96	69 / 94	55 / 88
steer, ECE	0.33	0.26	0.18
random, fail / ok	84 / 96	84 / 96	unparseable

88 tracks *whether this attempt succeeded*. Two further cuts of the data agree. Residualizing
 89 post-attempt probe scores on pre-attempt scores leaves substantial AUROC (0.59–0.87), so
 90 the post signal is attempt-specific rather than difficulty re-expressed. And the post-attempt
 91 direction transfers only weakly to the pre-attempt site. *Did I succeed?* and *Is this hard?* are
 92 largely distinct directions. Blind post-attempt confidence drops are also larger on failures
 93 than on successes.

94 **Generality: a substantially shared correctness direction.** If the model had one
 95 internal sense of “this is wrong,” a probe trained on one subject should work on another.
 96 It does: a probe trained in one domain and evaluated in another keeps most of its in-
 97 domain AUROC. In the original matrix, retention was about 87–90% (Qwen3.6 off-diagonal
 98 0.71 vs. diagonal 0.81), and sometimes lossless within QA (Qwen3.6 MMLU-Pro→MATH
 99 0.86 vs. 0.91). A fresh Mistral capture across all four domains, analysed with a corrected
 100 standardization step, gives 97% retention (Table 2: off-diagonal mean 0.69 vs. diagonal 0.71),
 101 with direction cosines of 0.34–0.82. GPQA is Mistral’s weakest domain (in-domain 0.57),
 102 and transfer into it is bounded accordingly. On Mistral the geometry anomaly does not
 103 recur: a MATH-trained probe reads geometry at 0.80, above geometry’s own in-domain 0.68,
 104 so the anti-transfer below is specific to Gemma-4.

105 Geometry is partly special on one model. On Gemma-4, geometry-trained probes *anti-*
 106 *transfer* to QA (0.31–0.39), while QA-trained probes work on geometry (about 0.70). Probes
 107 trained on the generative task can latch onto task-specific features; QA-trained probes find
 108 the general direction. Cosines of 0.34–0.82 and off-diagonal AUROC near 0.69 are consistent

with a shared correctness subspace rather than a single invariant axis, which is what a deployable readout needs and what a workspace account (below) predicts.

Causal access: the signal influences stated confidence. Reading shows the signal exists; steering shows it feeds the confidence the model reports. We steer at the confidence token, after the answer is written, so the answer cannot change. The direction is supervised: it is learned from labeled training examples, then applied without per-example labels at intervention time. On Mistral $\times$ MATH, as gain rises from 1 to 2 to 4, stated confidence on *failed* answers falls from 84 to 69 to 55 while correct answers hold near 90, and calibration error falls from 0.33 to 0.18 (Table 2, right). Dampening the signal *increases* overconfidence, so the dose-response runs both ways. This is evidence that the representation influences the report, not yet that the model consults it by default; mediation and ablation would be needed for that. These are the replication numbers, from a fresh 750-record capture with the random control scaled to the same per-record perturbation magnitude; the matched random direction leaves confidence unchanged at gain 2 and only degrades the output at gain 4. Replication depends on architecture. Gemma-4 shows the right specificity but is underpowered. GLM is null at the gains we tested; its stated confidence saturates near 100, exactly where the probe most exceeds $P(\text{True})$. We read this spectrum as the finding: *how much internal knowledge reaches speech varies by model*.

Expression: a label-free witness, and an architecture-gated gate. Trained probes invite an overfitting critique, so we added an unsupervised instrument: the Jacobian lens of the global-workspace account [3]. The lens is fit on generic text, with no task labels anywhere in its construction, and scores the same stored activations as a wrong-vs-correct disposition. On dense models it lands in probe territory (Mistral 0.76–0.82; Qwen2.5-14B, a dense within-family control, 0.88) and beats stated confidence with no labels. Two independent instruments find one direction. And because the lens reads through the model’s own output pathway, the signal it finds is *positioned to influence that pathway*; whether it is verbalizable as a concept is a further question we do not test. On the other architectures the lens itself fails (linear-attention hybrid 0.43, MoE 0.31) while the supervised probe still reads the signal in every model. The failure has a fingerprint. Transported through the averaged map, the lens’s “correct” and “wrong” readout vectors collapse to cosine 0.96 on MoE, against about 0.4 on dense models. An averaged map is only faithful when the routing barely varies per input, and both failing architectures vary theirs: a mixture of experts by which expert fires, and a gated linear-attention stack (48 of Qwen3.6’s 64 layers) by its input-dependent gates. Swapping that hybrid for a dense model of the same family flips the lens from 0.43 to 0.88. So the *label-free shortcut* is architecture-gated, and the signal is not. The open frontier is the gate itself. We want a direct per-input wiring-variance measurement, and to learn whether unexpressed knowledge on the failing architectures is available to the output pathway but suppressed, or inaccessible to it. The question is no longer *whether* models know. It is *what gates the knowing from the saying*.

3 Discussion

For reliable mathematical agents. Where a domain admits an exact verifier, as geometry does, use it. Where it does not, the internal signal is the best attempt-specific readout we found. Abstaining on the probe’s lowest-scoring outputs raises accuracy far more than abstaining on low stated confidence. In the cells where verbalized abstention fails outright, internal abstention still works. The steering result suggests stated confidence itself can be made honest by amplification rather than retraining.

Limitations. We read at the confidence token, not during the attempt, so the judgment is shown available when asked, not formed while solving; an answer-token read is the natural next control. A probe could also key on properties of failed derivations rather than a dedicated metacognitive variable: low token probability, hedging, or on MATH an early-layer channel next to the answer text; our surface baseline bounds this without closing it. Linear probes at a single token undercount nonlinear or multi-token signal. The causal arm is one clean model plus one underpowered and one null replication, and two MATH cells ran at

reduced n . Whether the knowing–saying gap is an artifact of preference tuning is open; a base-model arm would test it.

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A The elicitation protocol, and one record end to end

This appendix gives the exact prompts behind every record, then one complete record from the Mistral-Small-24B $\times$ MATH cell, summarised first and then given in full.

One record, Mistral-Small-24B $\times$ MATH. *What integer n satisfies $0 \leq n < 18$ and $n \equiv -11213141 \pmod{18}$?* The model sets the problem up correctly and gets $18 \times 622953 = 11213154$ right, then writes $r = -11213141 + 11213154 = 3$ and answers 3. That subtraction gives 13, which is the gold answer. Stated confidence: 90 before the attempt, 95 after it. Other readouts of the same attempt: P(True) 0.32, mean answer log-prob -0.05 , probe **TODO: OOF score pending**.

A single arithmetic slip inside a fluent derivation. Having just made the error, the model revises its stated confidence upward, while other readouts of that same attempt disagree with it. The rest of this appendix gives the prompts and the full transcript.

A.1 Prompts

Each record is four forward passes. A read is first taken at the last prompt token of the bare question, before any elicitation, then three turns are generated. The confidence turns run with thinking off so the read lands on a clean one-line reply; the attempt turn runs with thinking on for hybrid reasoners. The fixed **Confidence:** N format is what makes the read site content-neutral: the digit token has identical local context on every record, so layer-0 activations carry no signal by construction.

207 **System prompt (MATH).**

208 Solve the problem step by step. Put your final answer in `\boxed{}`.

209 **Turn 1, prospective confidence.** The problem is followed by:

210 Before answering, estimate how likely you are to answer this question
 211 correctly. Reply with EXACTLY one line and nothing else:
 212 Confidence: N
 213 where N is an integer from 0 (certain to be wrong) to 100 (certain to be
 214 correct). Do NOT answer yet.

215 **Turn 2, the attempt.** The turn-1 reply is left in context as the assistant message, then:

216 Now give your final answer inside `\boxed{}`.

217 **Turn 3, retrospective confidence.** The attempt is left in context, then:

218 Now assess the answer you just gave. On a scale of 0 to 100, how confident
 219 are you that it is correct? Reply with exactly one line and nothing else:
 220 Confidence: N

221 **P(True) baseline.** A separate teacher-forced branch off turn 2, no generation. Next-token
 222 mass on True versus False:

223 Consider your final answer above. Is it correct? Reply with exactly one word:
 224 True or False.

225 Grading is external and is never shown to the model, so the turn-3 revision is blind self-
 226 assessment.

227 **A.2 One record, verbatim**

228 Record `math_test_number_theory_753`, sample 2 of 5.

229 **Problem.** What integer n satisfies $0 \leq n < 18$ and $n \equiv -11213141 \pmod{18}$?

230 **Turn 1 reply.** Confidence: 90

231 **Turn 2 reply (complete).**

232 First, we need to find the remainder when -11213141 is divided by 18.
 233 1. Calculate $-11213141 \bmod 18$:

$$-11213141 \div 18 \approx -622952.2778$$

234 The integer part of the division is -622953 .

$$-11213141 = 18 \times (-622953) + r$$

235 where r is the remainder.
 236 2. Calculate the remainder r :

$$r = -11213141 - 18 \times (-622953)$$

237
$$r = -11213141 + 11213154$$

238
$$r = 3$$

239 Therefore,

$$-11213141 \equiv 3 \pmod{18}$$

240 So, the integer n that satisfies $0 \leq n < 18$ and $n \equiv -11213141 \pmod{18}$ is 3.

241 **Turn 3 reply.** Confidence: 95

242 **What happened.** The method is right and every intermediate is right: $18 \times 622953 =$
 243 11213154 checks out, and the model has written the correct subtraction on the line above its
 244 answer. It then evaluates $-11213141 + 11213154$ as 3 rather than 13. Gold is 13. Nothing
 245 in the presentation marks the slip. The derivation is well-formed, correctly structured, and
 246 confidently laid out, which is precisely why stated confidence rises from 90 to 95 across an
 247 attempt that failed.

248 **Readouts of this attempt.**

	readout	value
	stated confidence, prospective (turn 1)	90
	stated confidence, retrospective (turn 3)	95
249	P(True)	0.32
	mean answer log-probability	-0.05
	residual-stream probe, out-of-fold	TODO: pending
	graded outcome	incorrect

250 This is one record, chosen to illustrate the failure mode rather than to carry evidential weight;
 251 the cell-level numbers are in Table 1.

252 B Calibration beyond ECE, and cell-level multiplicity

253 ECE depends on binning and on sample composition, so we report two binning-free proper
 254 scoring rules alongside it. On Mistral $\times$ MATH (750 records, accuracy 0.648), the model’s
 255 retrospective stated confidence gives Brier 0.262 and NLL 1.470 against a 10-bin ECE of
 256 0.271. P(True) on the same records gives Brier 0.386: better ordered than stated confidence,
 257 worse calibrated in absolute terms. The steering results in Table 2 are reported as ECE
 258 because the steered runs stored per-gain aggregates rather than per-record confidences;
 259 recomputing Brier and NLL at each gain requires re-running the intervention and is the first
 260 thing we would add.

261 The cell-level bootstrap counts in Table 1 are uncorrected. With 16 cells, some individually
 262 significant results are expected under the null, so the cell counts should be read as descriptive
 263 and the aggregate direction of the effect, positive in 15 of 16 cells and never significantly
 264 negative, as the claim that carries weight. A hierarchical model pooling across cells with
 265 model and domain as random effects is the right test and is not yet run.